

Ollama

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Demo Environment

Development board: Jetson Orin series motherboard

Ollama is an open source tool that aims to simplify the deployment and operation of large language models, allowing users to use high-quality language models in local environments.

1. Large Language Model (LLM)

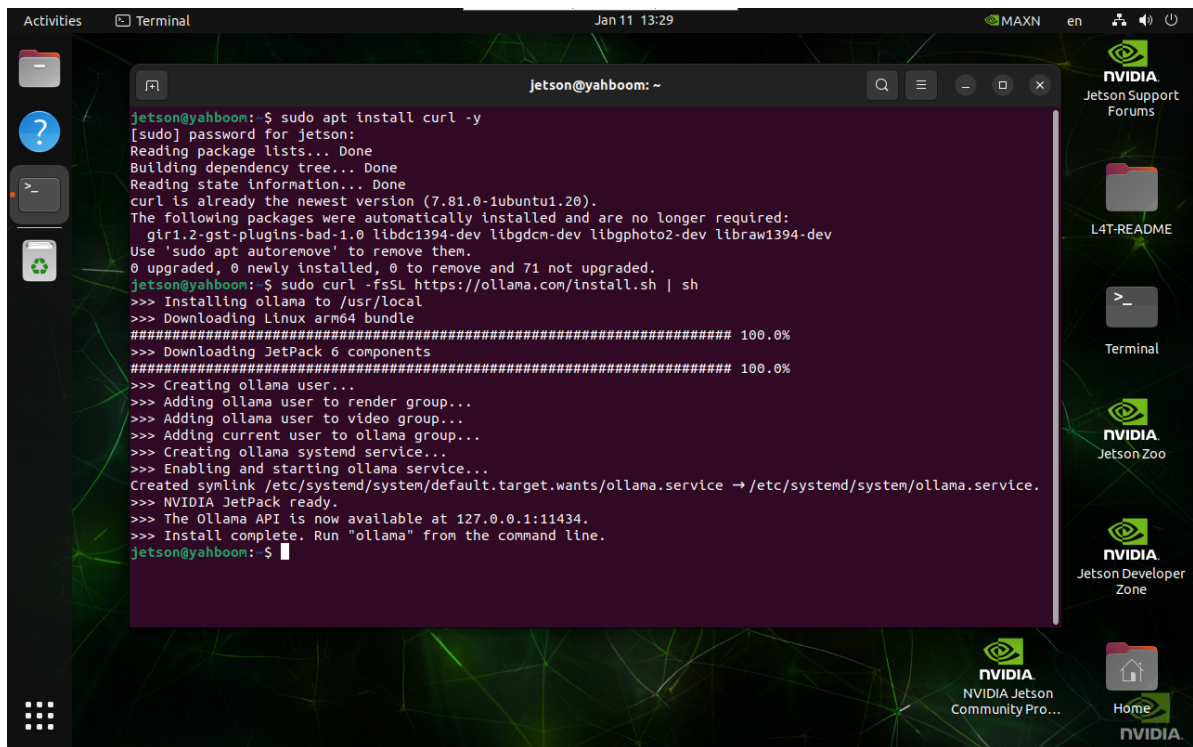
Large Language Models (LLM) are a type of advanced text generation system based on artificial intelligence technology. Its main feature is that it can learn and understand human language through large-scale training data and can generate natural and fluent text.

2. Ollama Installation

The tutorial demonstrates the use of scripts to install Ollama on the Jetson Orin series motherboard.

Script Installation

```
sudo apt install curl -y
sudo curl -fsSL https://ollama.com/install.sh | sh
```



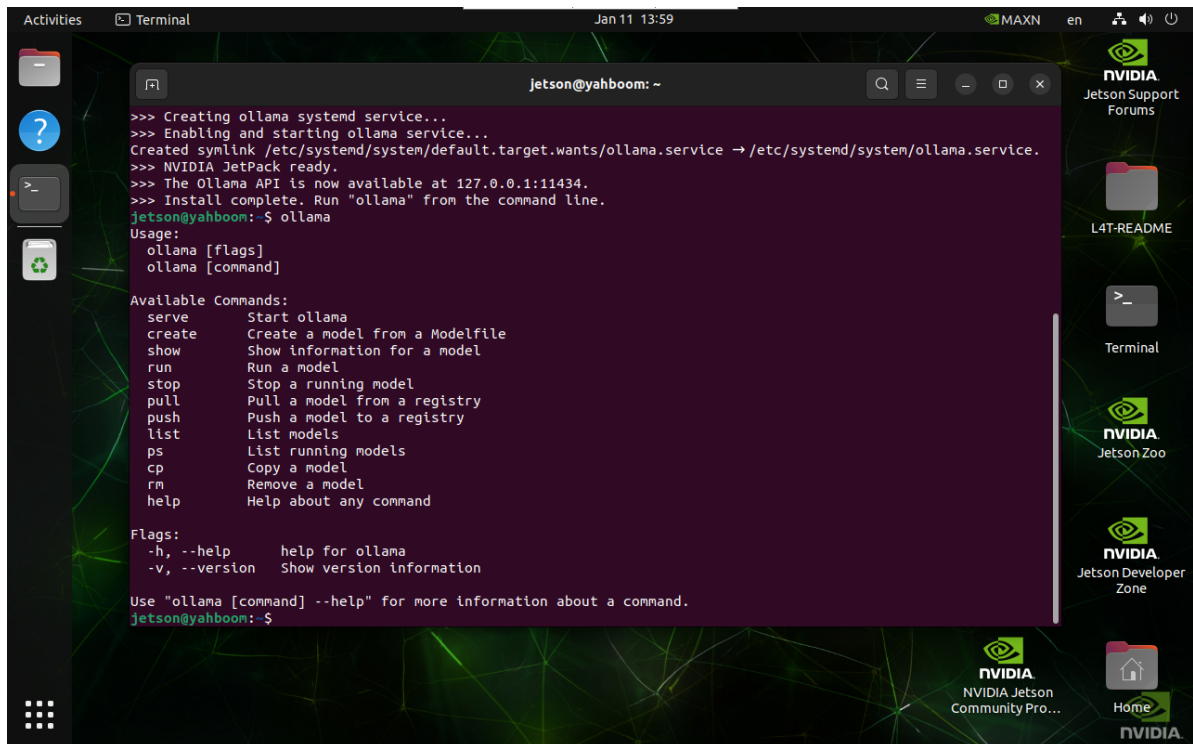
The terminal window shows the installation of Ollama on a Jetson device. The user runs `sudo apt install curl -y`, which installs curl. Then, the user runs `sudo curl -fsSL https://ollama.com/install.sh | sh`. The script performs several steps: installing ollama to /usr/local, downloading the Linux arm64 bundle, downloading JetPack 6 components, creating an ollama user, adding the user to the render and video groups, creating the ollama systemd service, enabling and starting the service, and finally, creating a symlink for the service. The installation is complete, and the Ollama API is available at 127.0.0.1:11434.

```
jetson@yahboom:~$ sudo apt install curl -y
[sudo] password for jetson:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (7.81.0-1ubuntu1.20).
The following packages were automatically installed and are no longer required:
  gir1.2-gst-plugins-bad-1.0 libdc1394-dev libgdcn-dev libgphoto2-dev libraw1394-dev
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 71 not upgraded.
jetson@yahboom:~$ sudo curl -fsSL https://ollama.com/install.sh | sh
>>> Installing ollama to /usr/local
>>> Downloading linux arm64 bundle
##### 100.0%
>>> Downloading JetPack 6 components
##### 100.0%
>>> Creating ollama user...
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
Created symlink /etc/systemd/system/default.target.wants/ollama.service → /etc/systemd/system/ollama.service.
>>> NVIDIA JetPack ready.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
jetson@yahboom:~$
```

The entire installation process takes a long time, please wait patiently!

3. Use Ollama

Type ollama in the terminal and you will see the prompt:



The terminal window shows the Ollama command-line interface. The user runs `ollama`, which displays the usage information and a list of available commands. The user can then run `ollama [command]` to perform various actions like starting, creating, showing, running, stopping, pulling, pushing, listing, and removing models.

```
jetson@yahboom:~$ ollama
Usage:
  ollama [flags]
  ollama [command]

Available Commands:
  serve    Start ollama
  create   Create a model from a Modelfile
  show     Show information for a model
  run      Run a model
  stop     Stop a running model
  pull     Pull a model from a registry
  push     Push a model to a registry
  list     List models
  ps       List running models
  cp       Copy a model
  rm       Remove a model
  help     Help about any command

Flags:
  -h, --help    help for ollama
  -v, --version  Show version information

Use "ollama [command] --help" for more information about a command.
jetson@yahboom:~$
```

Command	Purpose
ollama serve	Start ollama
ollama create	Create a model from a model file
ollama show	Show model information
ollama run	Run a model
ollama pull	Pull a model from a registry
ollama push	Push a model to a registry
ollama list	List models
ollama ps	List running models
ollama cp	Copy a model
ollama rm	Delete a model
ollama help	Get help information about any command

4. Ollama Uninstall

- Deleting a service

```
sudo systemctl stop ollama
sudo systemctl disable ollama
sudo rm /etc/systemd/system/ollama.service
```

- Deleting files

```
sudo rm $(which ollama)
```

- Deleting model and service users and groups

```
sudo rm -r /usr/share/ollama
sudo userdel ollama
sudo groupdel ollama
```

References

Ollama

Official website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>